Data-Driven Evaluation of School Infrastructure and Digital Access Factors Affecting Learning Outcomes Across Indian States

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Submitted By

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Domain

Data Analytics

Dataset Used

ASER 2024 State-Level Educational Indicators

Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, Statsmodels

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Abstract

Educational planners and policymakers often face the challenge of determining which interventions should receive priority in order to improve student learning outcomes. While traditional investments focus on infrastructure development and teacher availability, the rapid growth of digital technologies has introduced new factors that may significantly influence educational performance. This study aims to identify the educational and digital-access variables most strongly associated with student learning outcomes across Indian states and develop a data-driven framework for resource allocation.

The analysis was conducted using state-level indicators extracted from the Annual Status of Education Report (ASER) 2024. A consolidated dataset containing 22 Indian states and 30 educational indicators was constructed. Variables related to digital access, smartphone usage, educational activities, school infrastructure, playground availability, drinking water facilities, toilet facilities, and pupil-teacher ratio compliance were examined. Student reading performance at the Grade VIII level was selected as the primary outcome variable.

A combination of correlation analysis, Ordinary Least Squares (OLS) regression, and Random Forest feature importance techniques was employed to evaluate the relationship between educational inputs and learning outcomes. Correlation analysis revealed positive associations between reading performance and factors such as smartphone availability, smartphone usage, playground facilities, and pupil-teacher ratio compliance. Although the regression model exhibited limited explanatory power due to the relatively small sample size and the complexity of educational outcomes, machine learning-based feature importance analysis provided meaningful insights into the relative influence of different variables.

The results indicated that smartphone usage emerged as the most influential factor, accounting for approximately 40.09

Based on the analytical findings, a budget allocation model was developed to guide educational investments. The proposed framework recommends prioritizing digital learning initiatives and smartphone accessibility while continuing investments in school infrastructure and teacher availability. The study demonstrates the value of data-driven decision making in educational planning and highlights the increasing importance of digital inclusion in improving student learning outcomes across Indian states.

1 Introduction

Education is one of the most important drivers of social and economic development. Improving learning outcomes among school-going children remains a key objective of educational policy in India. Governments continuously invest in infrastructure development, teacher recruitment, educational technology, and welfare programs to improve the quality of education. However, identifying which interventions contribute most effectively to student learning remains a significant challenge.

In recent years, digital technologies have transformed the educational landscape. The increasing availability of smartphones and internet-based learning resources has enabled students to access educational content beyond traditional classrooms. At the same time, conventional factors such as school infrastructure, playground facilities, and pupil-teacher ratios continue to influence the learning environment and overall educational experience.

Understanding the relative importance of these factors is essential for effective policy formulation and resource allocation. Data-driven approaches provide an opportunity to objectively evaluate the relationship between educational inputs and student outcomes, thereby enabling evidence-based decision making.

This project uses state-level indicators from ASER 2024 to investigate the factors associated with student reading performance across Indian states. By analyzing both digital access indicators and school infrastructure variables, the study seeks to identify the most influential contributors to learning outcomes and propose a practical framework for educational budget allocation.

2 Problem Statement

Despite substantial investments in the education sector, learning outcomes continue to vary significantly across Indian states. Policymakers often face challenges in determining which interventions should receive higher priority in resource allocation decisions. While improvements in school infrastructure and teacher availability remain important, the rapid expansion of digital learning resources introduces new variables that may significantly affect educational outcomes.

This study aims to identify the factors most strongly associated with student reading performance and develop a data-driven framework for prioritizing educational investments.

3 Objectives

The primary objective of this study is to identify the factors most strongly associated with student learning outcomes across Indian states using data-driven analytical techniques.

The specific objectives are as follows:

1. To analyze educational indicators obtained from the ASER 2024 report.
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2. To examine the relationship between digital access variables and student learning outcomes.

3. To evaluate the influence of school infrastructure and pupil-teacher ratio compliance on educational performance.

4. To identify and rank the most important factors affecting Grade VIII reading ability.

5. To develop a data-driven budget allocation framework for educational investments based on analytical findings. “‘

4 Dataset Description

The dataset used in this study was constructed using state-level educational indicators extracted from the Annual Status of Education Report (ASER) 2024. Relevant information was manually collected and consolidated into a structured dataset suitable for statistical analysis.

The final dataset contains observations from 22 Indian states and includes a total of 30 variables representing learning outcomes, digital accessibility, school infrastructure, and compliance with educational standards.

The primary target variable selected for analysis was Grade VIII reading ability, measured as the percentage of students capable of reading at the Grade II level.

Table 1 summarizes the major categories of variables used in the study.

Table 1: Categories of Variables Used in the Study

Category	**Variables**
Learning Outcomes	Grade VIII Reading Ability
Digital Access	Smartphone Access at Home, Smartphone Usage, Educational Activity Online
School Infrastructure	Playground Availability, Drinking Water Availability, Toilet Facilities, Girls' Toilet Facilities
Educational Standards	PTR Compliance, CTR Compliance

The dataset was examined for completeness and consistency prior to analysis. All variables were converted to numeric format wherever applicable. Missing value analysis revealed that no missing observations were present in the dataset. Consequently, no imputation or data reconstruction techniques were required.

The resulting dataset provided a clean and reliable foundation for statistical analysis and machine learning-based feature evaluation.

5 Methodology

A multi-stage analytical framework was adopted to investigate the relationship between educational indicators and student learning outcomes. The methodology consisted of data preprocessing, exploratory analysis, statistical modeling, and machine learning-based feature evaluation.

The overall workflow adopted in this study consisted of the following stages:

1. Data Collection and Consolidation
2. Data Cleaning and Validation
3. Correlation Analysis
4. Regression Modeling
5. Feature Importance Analysis
6. Budget Allocation Framework Development

5.1 Data Collection and Preprocessing

Educational indicators were extracted from multiple tables published in the ASER 2024 report. Relevant variables related to learning outcomes, digital accessibility, school infrastructure, and educational standards were selected and consolidated into a single dataset.

The dataset was cleaned using Python's Pandas library. Variable names were standardized, data types were converted into numerical format, and the dataset was inspected for inconsistencies and missing values. Since no missing observations were identified, no imputation procedures were required.

5.2 Correlation Analysis

Pearson correlation analysis was performed to identify linear relationships between educational indicators and Grade VIII reading ability. Correlation coefficients were used to measure both the strength and direction of associations between variables.

The resulting correlation matrix provided an initial understanding of which variables exhibited positive or negative relationships with student learning outcomes and helped identify candidate predictors for subsequent modeling stages.

5.3 Ordinary Least Squares Regression

Ordinary Least Squares (OLS) regression was employed to evaluate the combined influence of selected predictor variables on Grade VIII reading performance.

The regression model was used to estimate the magnitude and direction of the relationship between educational indicators and learning outcomes while controlling for the influence of other variables included in the model. Regression coefficients, statistical significance measures, and goodness-of-fit metrics were examined to assess model performance.

5.4 Random Forest Feature Importance Analysis

To overcome limitations associated with linear modeling and to obtain a robust ranking of predictor variables, a Random Forest Regression model was implemented using the Scikit-Learn library.

Random Forest models are capable of capturing non-linear relationships between variables and often provide more reliable estimates of variable importance in complex datasets. Feature importance scores generated by the Random Forest model were used to quantify the relative contribution of each variable to learning outcomes.

The resulting importance scores formed the basis for the educational budget allocation framework proposed in this study.

5.5 Budget Allocation Framework

A budget allocation framework was developed using the feature importance scores generated by the Random Forest model. Variables with higher importance scores were assigned a greater proportion of the hypothetical educational investment budget.

This approach provides a practical mechanism for translating analytical findings into actionable policy recommendations and resource allocation strategies.

5.6 Software and Tools

The analysis was performed using Python and the following open-source libraries:

- Pandas for data manipulation and preprocessing
- NumPy for numerical computation
- Matplotlib for data visualization
- Seaborn for statistical visualization
- Statsmodels for regression analysis

- Scikit-Learn for Random Forest modeling

All experiments and analyses were conducted within a Jupyter Notebook environment using Visual Studio Code.

6 Results and Analysis

This section presents the findings obtained through correlation analysis, regression modeling, and Random Forest feature importance evaluation. The objective of the analysis was to identify the educational and digital-access factors most strongly associated with Grade VIII reading performance across Indian states.

6.1 Correlation Analysis

A correlation matrix was generated to examine the relationships between educational indicators and student learning outcomes. Positive correlation values indicate that an increase in a variable is associated with improved reading performance, whereas negative values indicate the opposite.

The analysis revealed that smartphone access at home, smartphone usage, playground availability, and pupil-teacher ratio compliance exhibited positive associations with Grade VIII reading ability.

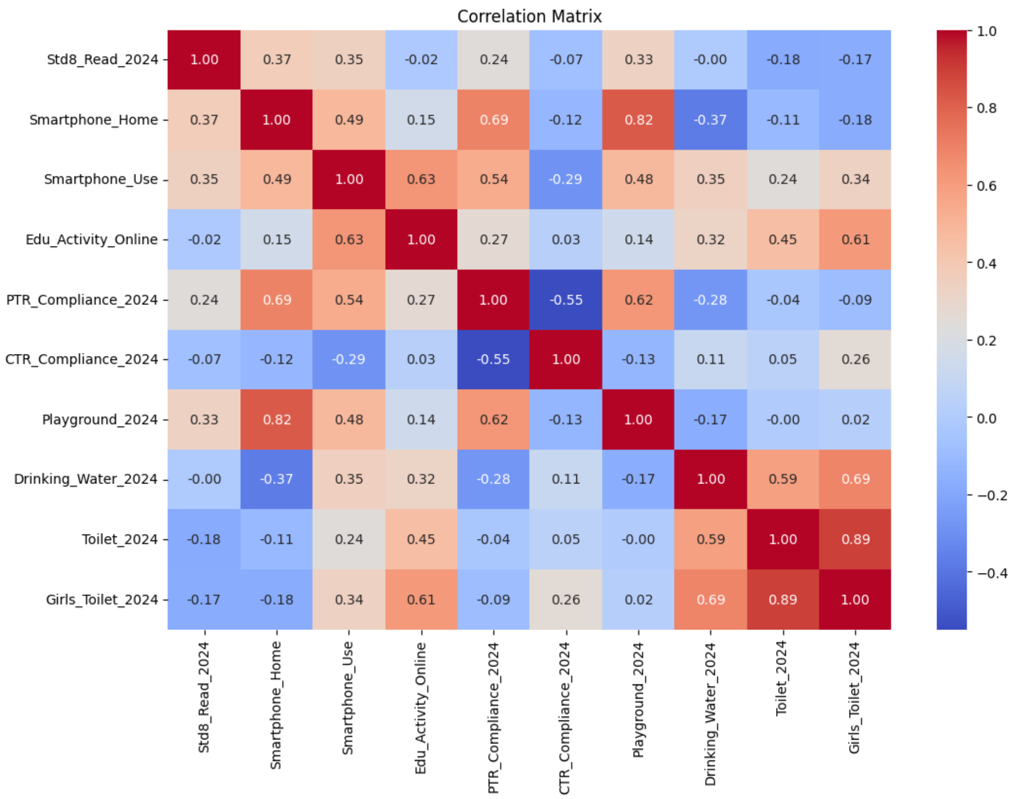

Figure 1: Correlation Matrix of Educational Indicators

Among the selected variables, smartphone access at home demonstrated a correlation coefficient of approximately 0.37 with reading performance, while smartphone usage exhibited a correlation of approximately 0.35. Playground availability also showed a moderate positive association.

These findings suggest that digital accessibility and learning opportunities beyond the classroom may play an important role in improving educational outcomes.

6.2 State-wise Learning Performance

To understand variations in learning outcomes across India, the states were ranked according to Grade VIII reading performance.

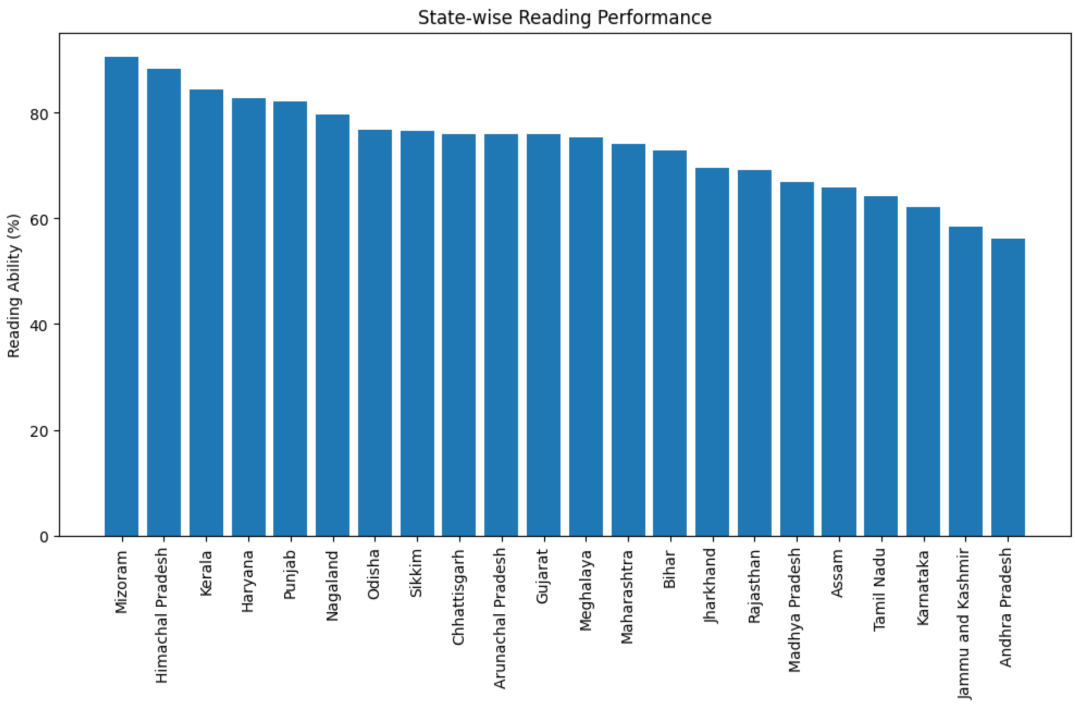

Figure 2: State-wise Reading Performance

The analysis revealed substantial differences among states. Mizoram achieved the highest reading performance score of 90.7

Such variations highlight the importance of identifying best practices from high-performing states and applying similar interventions in regions exhibiting lower educational outcomes.

6.3 Regression Analysis

Ordinary Least Squares (OLS) regression was performed to evaluate the combined influence of multiple educational indicators on Grade VIII reading ability.

The regression model produced a relatively low explanatory power, with an R^2 value of approximately 0.18. This indicates that while the selected variables contribute to learning outcomes, a substantial proportion of variation is influenced by factors not included in the dataset.

Several reasons may explain this result:

- The dataset contains observations from only 22 states.
- Learning outcomes are influenced by socioeconomic conditions, parental education, and regional factors not included in the study.
- Educational performance is inherently complex and cannot be fully explained by a limited set of indicators.

Despite these limitations, the regression analysis provided useful insights regarding the direction of relationships between variables and learning outcomes.

6.4 Random Forest Feature Importance Analysis

To obtain a more robust ranking of influential variables, a Random Forest Regression model was employed.

Feature importance scores generated by the model identified the relative contribution of each factor toward predicting reading performance.

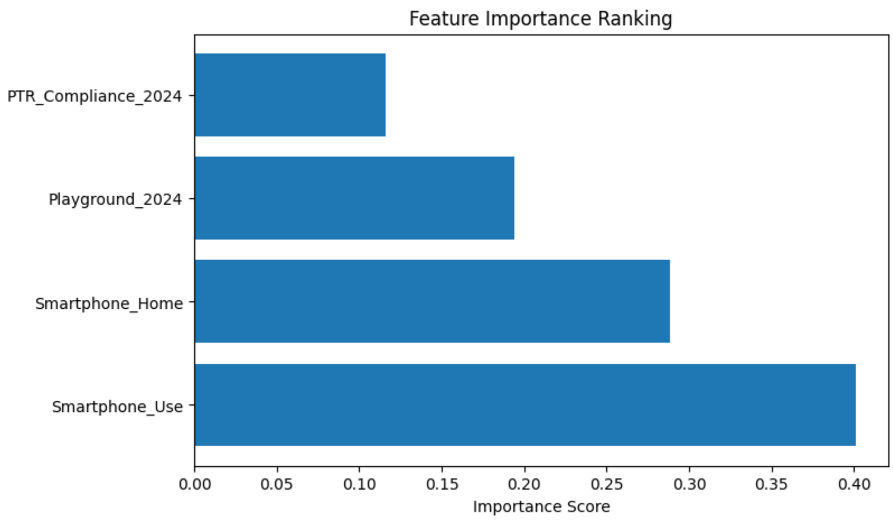

Figure 3: Feature Importance Ranking

The results revealed that smartphone usage emerged as the most influential variable, followed by smartphone access at home, playground availability, and pupil-teacher ratio compliance.

Table 2 summarizes the obtained feature importance scores.

Table 2: Feature Importance Scores

Factor	**Importance (%)**
Smartphone Usage	40.09
Smartphone Access	28.85
Playground Availability	19.43
PTR Compliance	11.63

The findings indicate that digital-access indicators collectively account for nearly 69

6.5 Effect Size Analysis

The correlation coefficients between Grade VIII reading performance and the selected educational indicators are presented in Table 3. These values provide an estimate of the relative strength of association between each factor and reading outcomes.

Table 3: Effect Size Indicators

Variable	Correlation Coefficient
Smartphone Access	0.37
Smartphone Usage	0.35
Playground Availability	0.33
PTR Compliance	0.24

The results indicate that smartphone access and smartphone usage exhibit the strongest positive associations with reading performance, followed by playground availability and pupil-teacher ratio compliance.

7 Budget Allocation Framework

The feature importance scores generated by the Random Forest model were used to design a data-driven educational budget allocation framework. The objective of this framework is to demonstrate how analytical findings can be translated into practical policy recommendations for educational investment planning.

The underlying assumption is that factors exhibiting greater influence on learning outcomes should receive proportionally higher investment priority. Therefore, the percentage contribution of each feature was directly converted into a recommended budget allocation percentage.

Figure 4 illustrates the proposed allocation strategy.

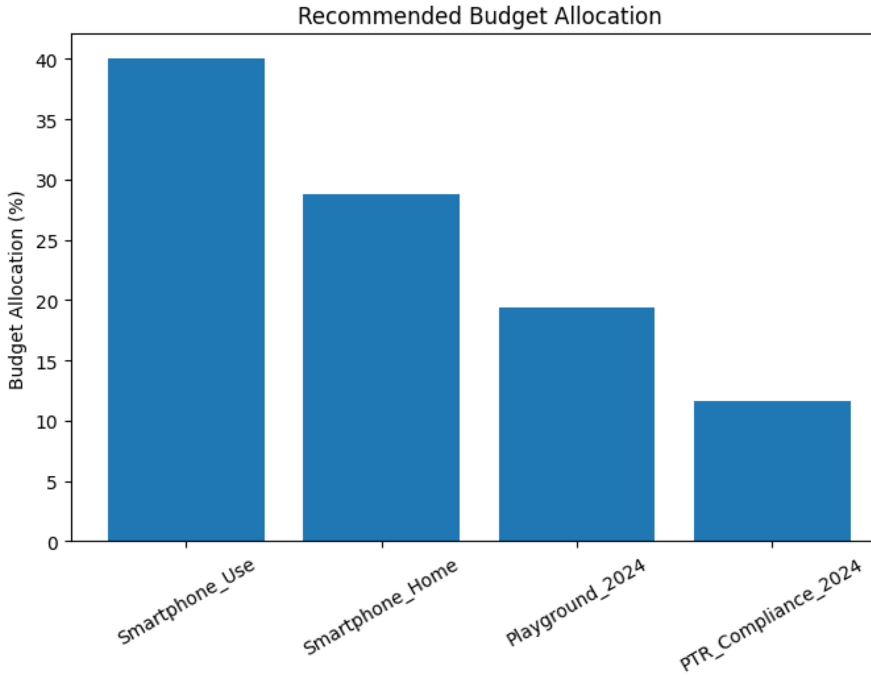

Figure 4: Recommended Budget Allocation Based on Feature Importance

The proposed allocation percentages are summarized in Table 4.

Table 4: Recommended Budget Allocation

Investment Area	**Allocation (%)**
Smartphone Usage Programs	40.09
Smartphone Accessibility	28.85
Playground Development	19.43
PTR Compliance Improvement	11.63

The results suggest that investments focused on improving digital accessibility and effective technology utilization may yield substantial educational benefits. Nearly 69% of the recommended budget allocation is directed toward digital-access initiatives, reflecting the strong influence of smartphone availability and smartphone usage on student reading performance.

At the same time, infrastructure-related interventions such as playground development and teacher availability remain important contributors to learning outcomes. Therefore, a balanced investment strategy combining digital inclusion and traditional educational support mechanisms is recommended.

8 Conclusion

This study investigated the relationship between educational infrastructure, digital accessibility, and student learning outcomes across Indian states using state-level indicators extracted from the ASER 2024 report.

A consolidated dataset containing 22 states and 30 educational variables was developed and analyzed using correlation analysis, Ordinary Least Squares regression, and Random Forest feature importance techniques. Grade VIII reading ability was selected as the primary learning outcome indicator.

The analysis revealed positive associations between reading performance and factors such as smartphone accessibility, smartphone usage, playground availability, and pupil-teacher ratio compliance. Random Forest feature importance analysis identified smartphone usage as the most influential predictor of reading performance, followed by smartphone accessibility, playground facilities, and PTR compliance.

Based on these findings, a budget allocation framework was developed to support evidence-based educational planning. The framework recommends prioritizing investments in digital learning access while maintaining support for school infrastructure and teacher availability.

8.1 Limitations

The present study is based on observational state-level data and therefore identifies statistical associations rather than causal relationships. While the proposed budget allocation framework prioritizes factors with stronger observed relationships to reading performance, the actual magnitude of improvement cannot be directly estimated from the available dataset.

The proposed allocation framework is intended to improve learning outcomes by prioritizing high-impact intervention areas. Actual improvement percentages would require field implementation, longitudinal monitoring, and experimental evaluation to establish causal effects.

Although the study is limited by the relatively small number of state-level observations, it demonstrates the usefulness of data analytics and machine learning techniques in identifying educational priorities and supporting policy decision-making.

9 Future Work

Several opportunities exist for extending this work in future studies:

- Incorporating district-level educational data to significantly increase sample size and improve model reliability.
- Including socioeconomic indicators such as household income, parental education levels, internet connectivity, and rural-urban classification.

- Applying advanced machine learning algorithms such as XGBoost, CatBoost, and Gradient Boosting Regression for improved predictive performance.
- Developing an interactive dashboard that allows policymakers to simulate educational investment scenarios and evaluate expected outcomes.
- Expanding the framework to include additional learning outcome measures such as mathematics proficiency, attendance rates, and school completion rates.

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